

LAB EXAM

Problem 1: By using DHT Sensor check Humidity and Temperature Using Arduino IDE (Integrated Development Environment) on Serial Monitor and also check it in the node red dashboard.

- **Using components:**

- ✓ Arduino UNO R3
- ✓ DHT 22 Sensor
- ✓ 3 jumper wire
- ✓ USB 2.0(Type-A to Type-B)

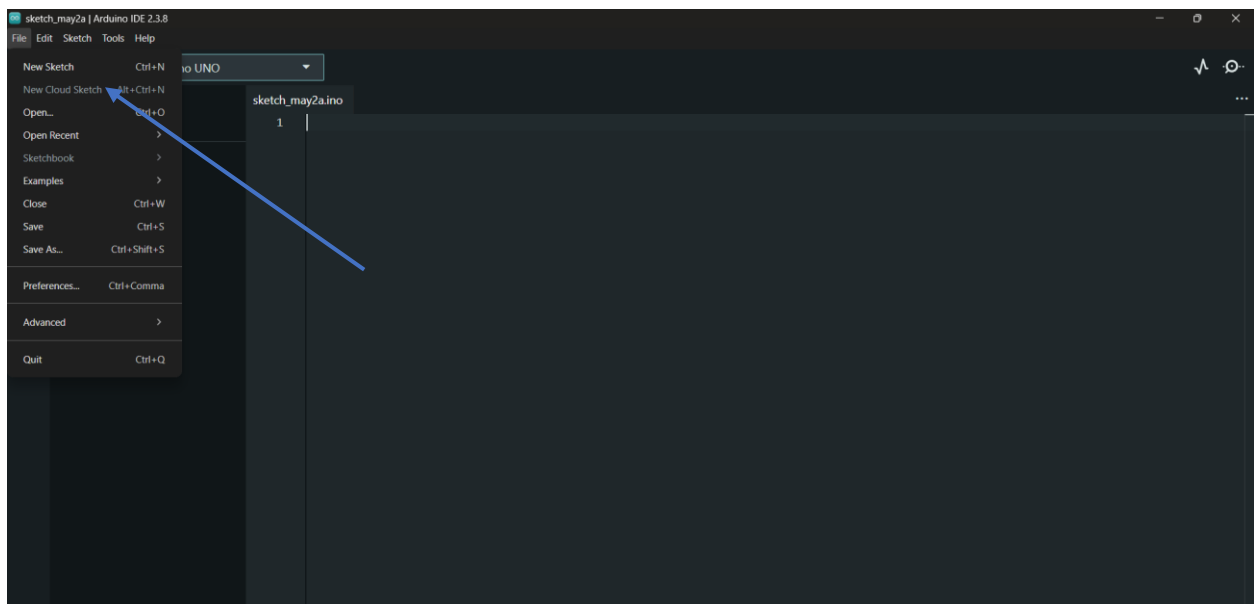
- **Arduino IDE:**

Arduino IDE is a Development Environment where to write, compile and upload the code into the Arduino UNO R3 board.

in simple word It is the text editor for this hardware.

Step-1:

Open new sketch file.



Step 2:-

write the code for the dht22 sensor in (C++)

Code:

```

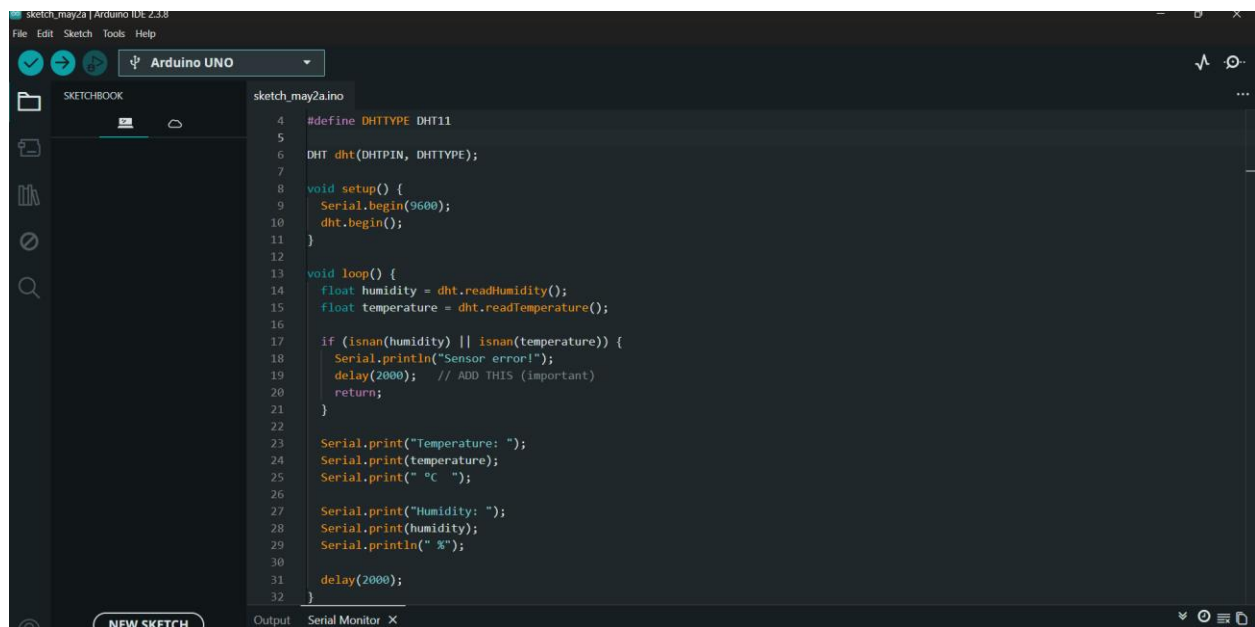
#include <DHT.h>
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

void setup () {
  Serial. Begin(9600);
  dht.begin();
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Sensor error!");
    delay(2000); // ADD THIS (important)
    return;
  }
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" °C ");

  Serial.print("Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");
  delay(2000);
}

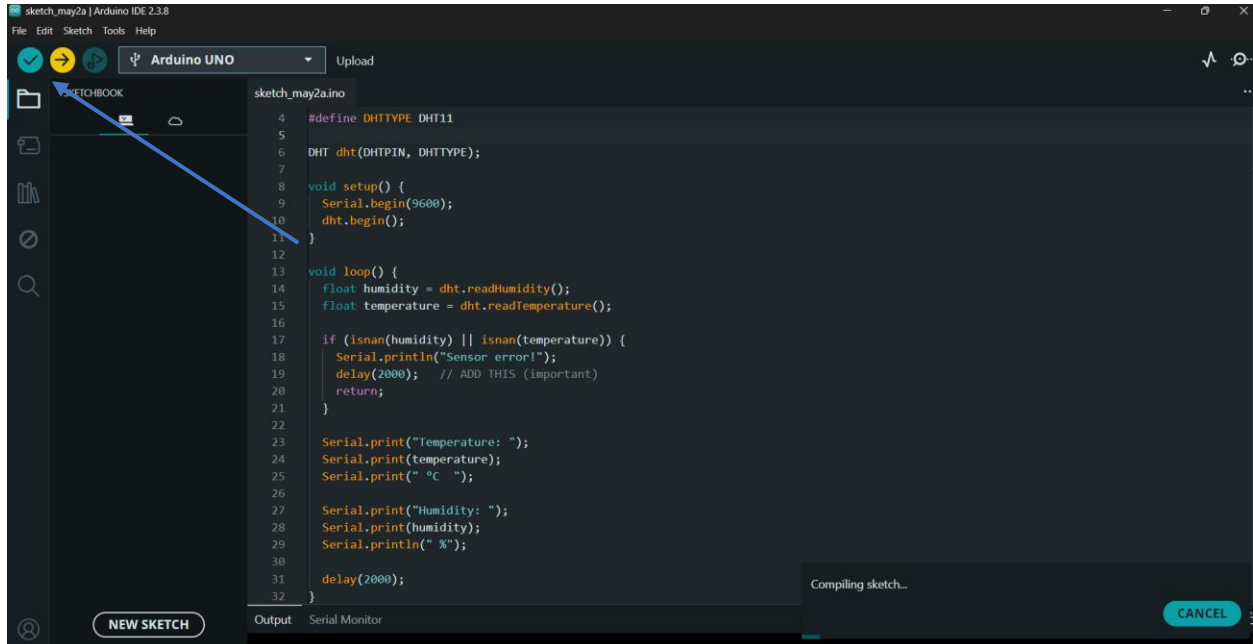
```



Step 3:-

Before compile the code u just select the board on Arduino uno box and see which port it will use to connect with your system.

then compile & upload it into Arduino board.



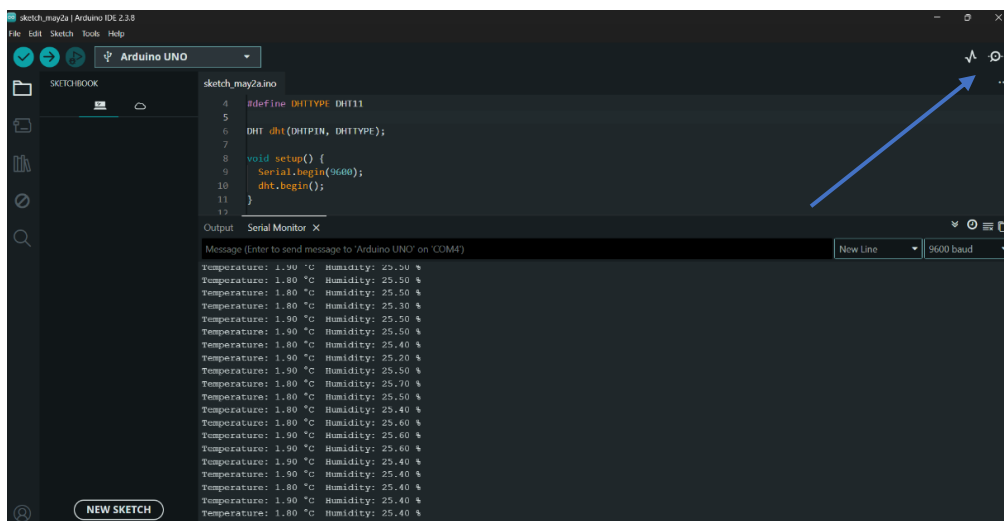
Verifying



Uploading into
Arduino Board

Step 4:-

Open the serial monitor. See result.



Here in sensor some kind of error to show the correct temperature in Serial monitor.

Code Output:

```
Temperature: 1.60 °C Humidity: 3.00 %  
Temperature: 1.60 °C Humidity: 3.00 %  
Temperature: 1.60 °C Humidity: 27.40 %  
Temperature: 1.60 °C Humidity: 27.50 %  
Temperature: 1.60 °C Humidity: 27.50 %  
Temperature: 1.60 °C Humidity: 27.50 %  
Temperature: 1.60 °C Humidity: 3.00
```

Step 5: -

After Successfully code running then we try in node red dashboard to show temperature and humidity in graphically.

- **Node red:**

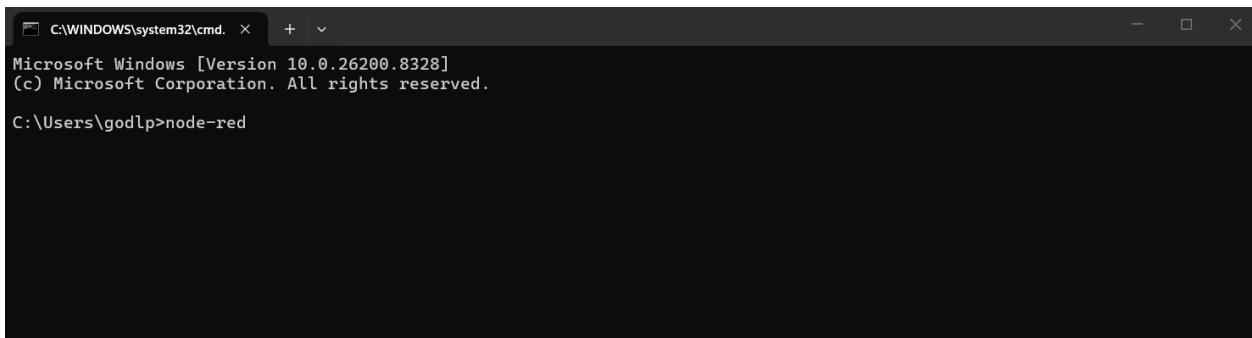
Where Arduino ide is a text editor for Arduino microcontroller but node red is a visual representation for showing any temp, humidity etc.

It is an opensource platform which is created by IBM and it is a flow-based programming tool.

It allows u to “Wire together” hardware devices and showing the temp, humidity on browser based Ui or editor.

Press windows + R (type CMD in box)

Open command prompt and type this command → (node-red)



```
C:\WINDOWS\system32\cmd. x + v  
Microsoft Windows [Version 10.0.26200.8328]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\godlp>node-red
```

Step 6: -

Before writing this command, u have installed the node red application in your windows. If u is not installed node red in your system below, I give a link how to install the node red application using this command prompt(cmd).

Node red Install: - <https://nodered.org/docs/getting-started/windows>

After Node red install u have install another one package→(Node JS) here for showing your temperature or humidity in graphically.

Node JS Install: - <https://nodejs.org/en/download/package-manager>

When u complete those two installations then u start the node red using the command

→(node-red) in cmd.

```
10 May 09:02:51 - [info] Node-RED version: v4.1.8
10 May 09:02:51 - [info] Node.js version: v24.15.0
10 May 09:02:51 - [info] Windows_NT 10.0.26200 x64 LE
10 May 09:02:53 - [info] Loading palette nodes
10 May 09:02:53 - [info] Dashboard version 3.6.6 started at /ui
10 May 09:02:53 - [info] Settings file : C:\Users\godlp\.node-red\settings.js
10 May 09:02:53 - [info] Context store : 'default' [module=memory]
10 May 09:02:53 - [info] User directory : \Users\godlp\.node-red
10 May 09:02:53 - [warn] Projects disabled : editorTheme.projects.enabled=false
10 May 09:02:53 - [info] Flows file : \Users\godlp\.node-red\flows.json
10 May 09:02:53 - [warn]

-----
Your flow credentials file is encrypted using a system-generated key.

If the system-generated key is lost for any reason, your credentials
file will not be recoverable, you will have to delete it and re-enter
your credentials.

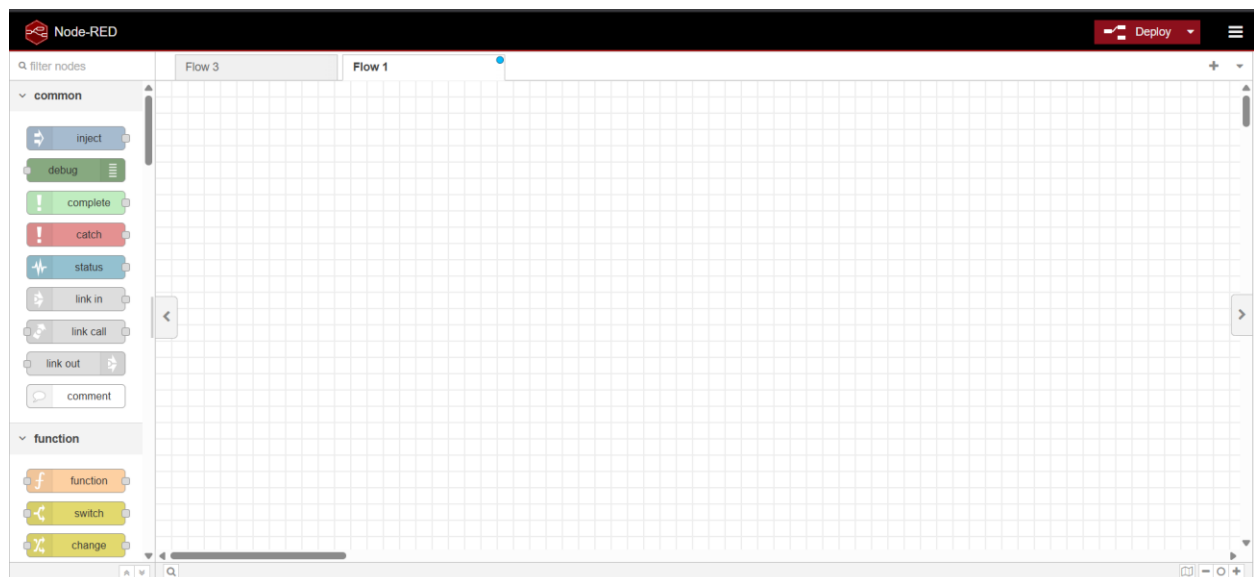
You should set your own key using the 'credentialSecret' option in
your settings file. Node-RED will then re-encrypt your credentials
file using your chosen key the next time you deploy a change.
-----
10 May 09:02:53 - [warn] Encrypted credentials not found
10 May 09:02:53 - [info] Server now running at http://127.0.0.1:1880/
10 May 09:02:53 - [info] Starting flows
10 May 09:02:53 - [info] Started flows
10 May 09:02:53 - [info] [serialconfig:58430f06.f23e4] serial port COM4 opened at 9600 baud 8N1
```

Step 7: -

In given picture u see one link was created (<http://127.0.0.1:1880/>).

Then press (CTRL+ Right click) with mouse pointer on its).

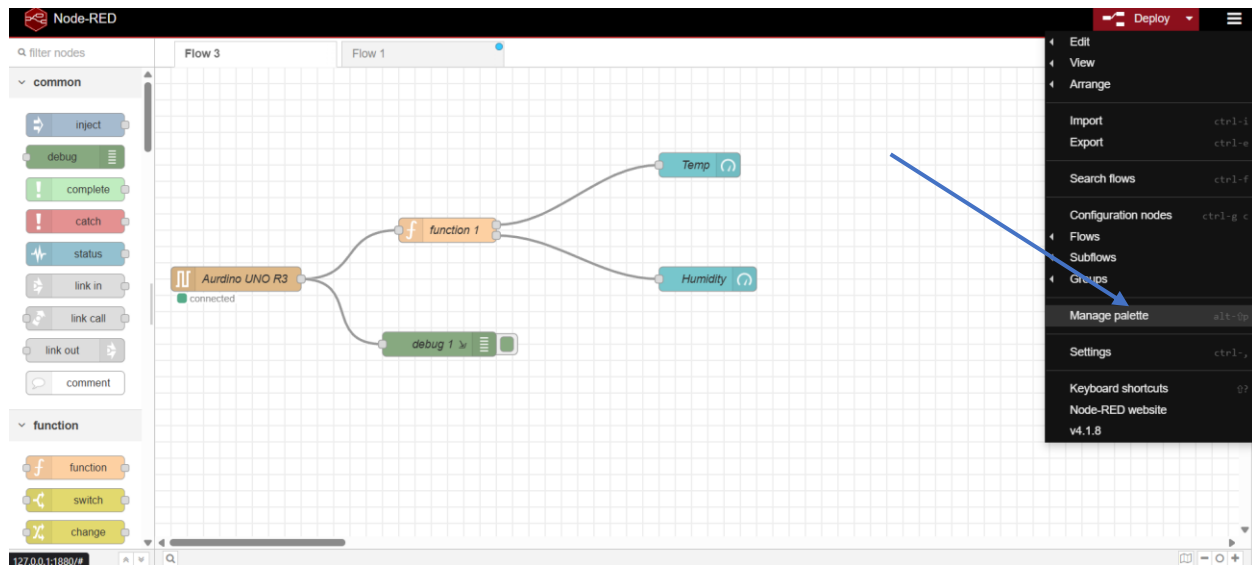
Then it will direct connect to the node red server and automatically open it on your browser. Like this:



After opening this type of interface here u install some library to create a graph or flow to connect with Arduino board.

Step 8: -

Next you just go to right corner and see the three dot or it is named as (Hamburger Menu) click on it and next u go to manage pallets section.



Step 9: -

After clicking on its u see at right side the install option then click on it and install these library's:

1. Node-red
2. Node-red-dashboard
3. Node-red-node-serial port

Then u just back it and use the filter sections to find these nodes:

1. Serial in
2. Function
3. Debug
4. Gauge (here u take two gauge one is temp and another one is humidity)

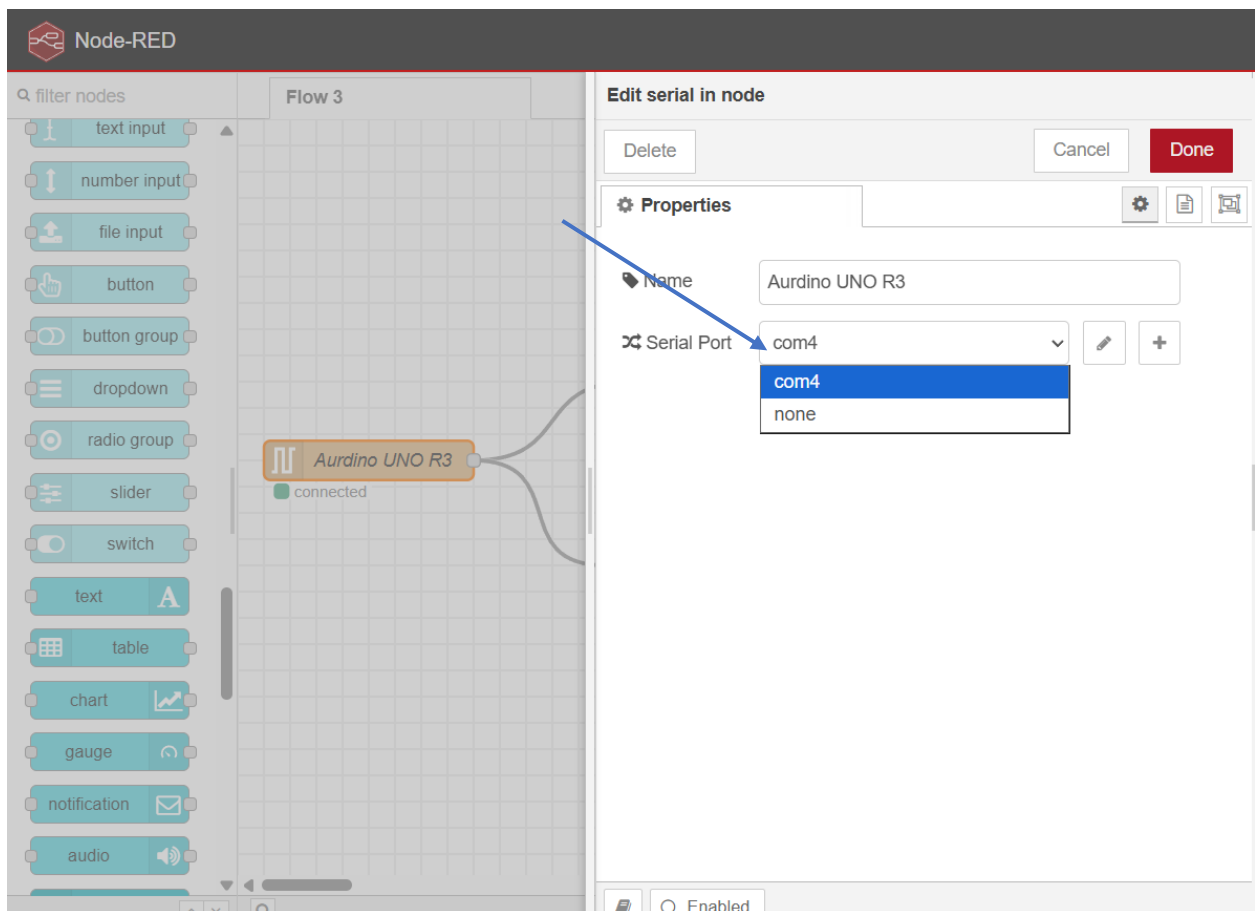
After taking these nodes then connect them giving below instructions:

serial in node → function node

Serial in node → debug node

Function node → gauge node (here two-gauge node, so u just use two connectors to connect them).

After connection u see that I give a same flow in given picture.



Then u just close the serial monitor In Arduino IDE.

In next step here u double tap on (serial in) node and set the port that Arduino will use to connect with your system. Like (COM4, COM5, COM6, etc.).

Step 10:-

After selecting the port then u just write the name of node (any name u have) then done it and move to the function node and double tap on it and write this code in (on message section).

Code:

```
// 1. Convert Buffer to String
```

```
let rawData = msg.payload.toString();
```

```
// 2. Use a Regular Expression to find only the numbers (and decimals)
```

```
// This extracts "25.5" out of "Temp: 25.5"
```

```
let numbers = rawData.match(/[-+]?[0-9]*\.[0-9]+/g);
```

```
if (numbers) {
```

```
    // numbers[0] is the first number found (Temp)
```

```
// numbers[1] is the second number found (Humidity)

let msg1 = { payload: parseFloat(numbers[0]) };

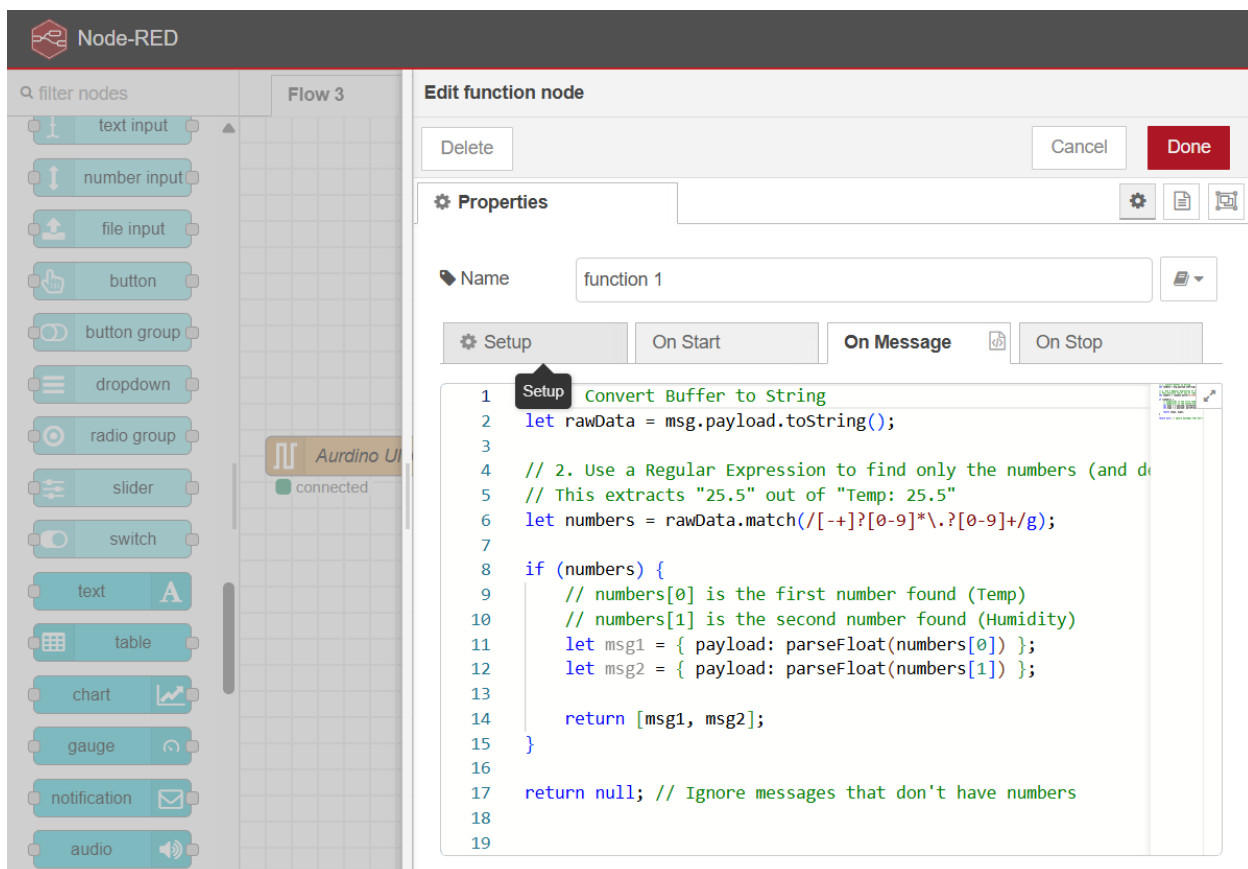
let msg2 = { payload: parseFloat(numbers[1]) };

return [msg1, msg2];

}

return null; // Ignore messages that don't have numbers
```

Step 11:-



After write the code just done it and move to the two gauge node and double tap on it and set these details :

For temperatre:

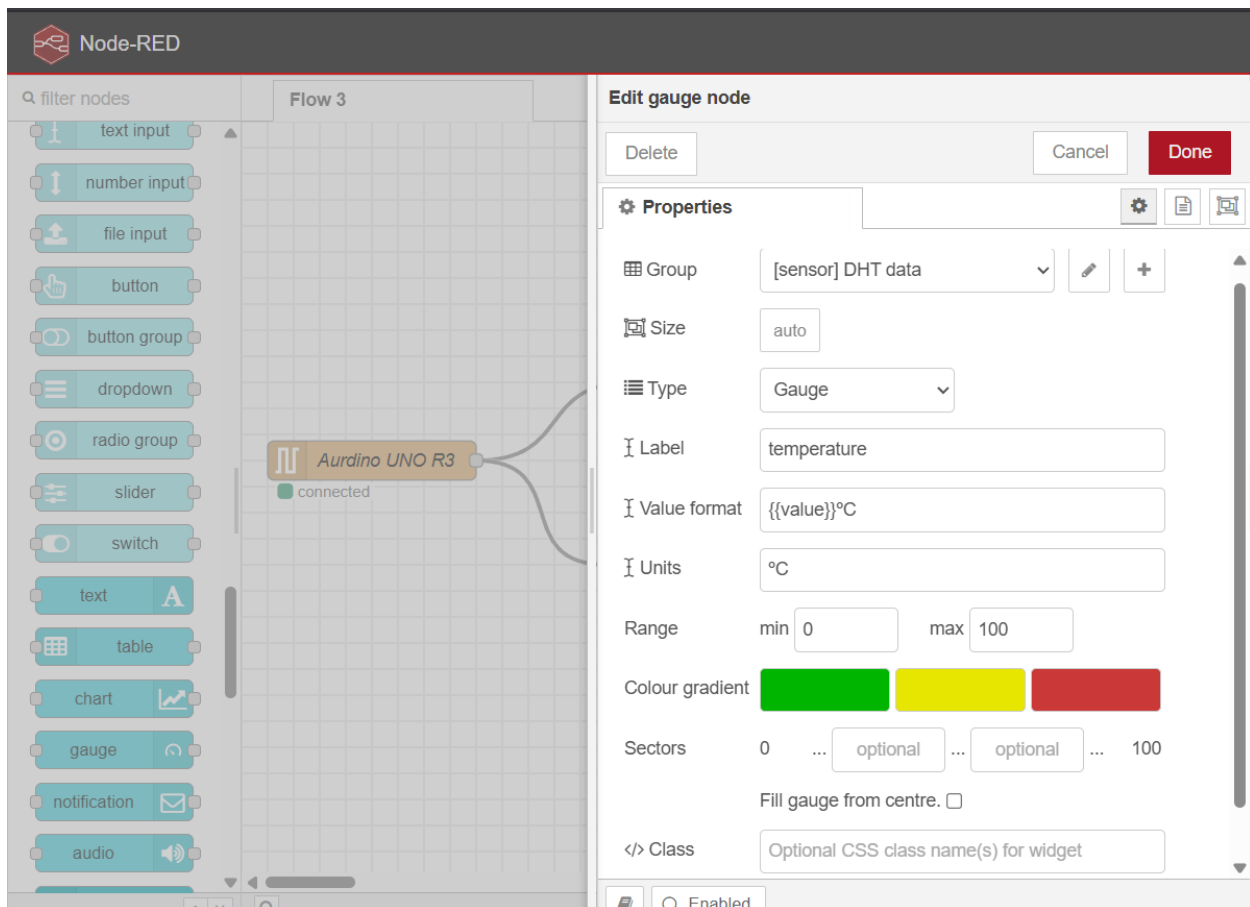
- Group-DHT sensor
- Label-temperature
- Label unit-(symbol: °C)
- Range min-0, max-100
- Name-temp

For humidity:

Same as temperature but details different.

- a. Group-DHT sensor
- b. Label-humidity
- c. Label unit-(symbol: %)
- d. Range min-0, max-100(it is same for two section nodes)
- e. Name-humidity

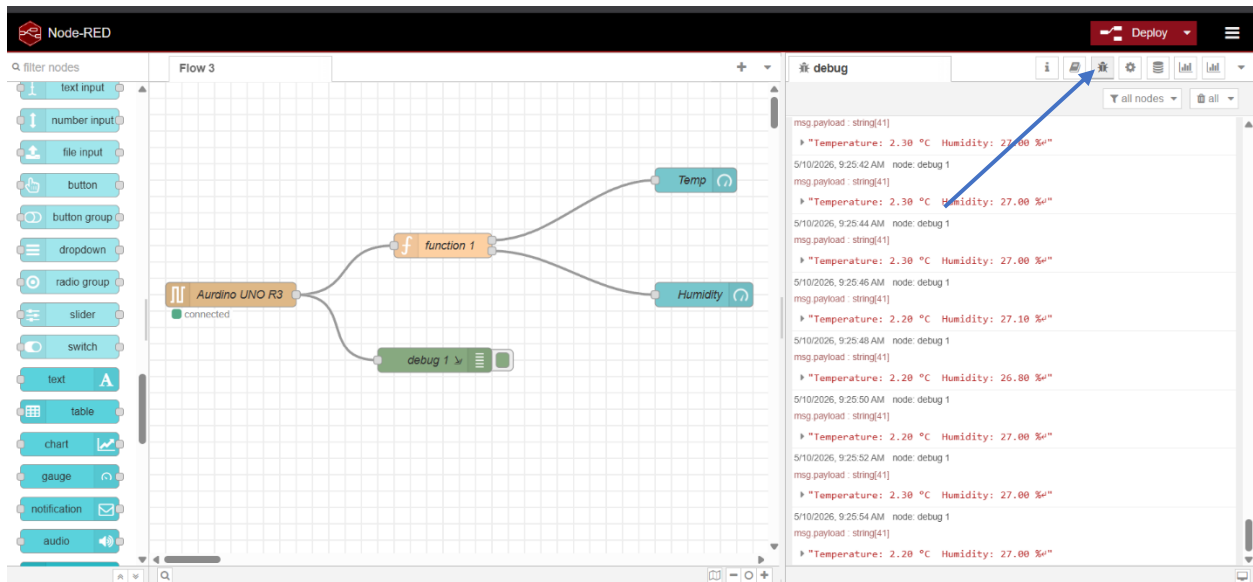
So here I give just one picture of that format details (temperature)



Here u see in group section the pencil symbol is to edit the dashboard group name and plus symbol (+) is to add a dashboard group name.

Step 12: -

Then u done it and click the deploy section on right section. After deploy it into your Arduino board see the resultor output of the flow in debug section.



See the output:

msg.payload : string [41]

"Temperature: 1.20 °C Humidity: 25.50 %"

Step 13: -

When successfully show the output then u just copy the dashboard port that I given below and paste it into your browser where u open the red node server.

Port: **http://localhost:1880/ui** (just copy it and paste it on your respective browser then it will open this type of interface.

Output:

